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Dynamic Pricing and Inventory Control of Substitute Products

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We study dynamic pricing and inventory control of substitute products for a retailer who faces a long supply lead time and a short selling season. Within a multinomial logit model of consumer choice over substitutes, we develop a stochastic dynamic programming formulation and derive the optimal dynamic pricing policy. We prove that dynamic pricing converges to static pricing as inventory levels of all variates approach the number of remaining selling periods (assuming at most one customer arrival within each period). Our extensive numerical study of the effects of time and inventory depletion on the optimal pricing reveals two fundamental underlying driving forces of the complex price behavior: the level of inventory scarcity and the quality difference among products. We also compare the performance of three restricted pricing strategies: static, unified dynamic, and mixed dynamic pricing. We find that full-scale dynamic pricing is of great value in the presence of inventory scarcity, and initial inventory decisions are quite robust in the pricing scheme employed in the selling season. Based on the above insights, we propose a computationally efficient approach to the initial inventory decision, which delivers close-to-optimal inventory levels for all testing cases.

Key words: dynamic pricing; inventory control; substitute products; multinomial logit model
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1. Introduction

1.1. Motivation and Overview of Contributions

Consumers typically choose among a variety of competing goods based on criteria such as features, style, quality, price, and availability. As advances of information technology allow firms to collect large amounts of data on recent sales and consumer behavior (e.g., point-of-sale data, online orders, tracking of consumer online searches prior to a purchase, etc.), smart data mining and effective information management processes help them turn it into a wealth of information on consumer preferences, price sensitivity, variety-seeking behavior, and market trends. Effective retailers move beyond a passive management of inventory and supply conditions in reaction to realized demand and actively influence consumer purchase decisions through smart assortment, real-time monitoring and displaying of inventories, and dynamic promotion and pricing decisions. Such dynamic demand management strategies improve the effective matching of supply and demand, allow for both improved efficiency and responsiveness of supply chains, and lead to substantially better returns on inventory investment. Companies that follow such practices are characterized as "AAA" world-class supply chain practitioners (see Lee 2004); in their ranks are companies such as Seven-Eleven Japan Co, Zara, Amazon, and Dell.

In this paper, we study the dynamic pricing and inventory control of differentiated (along one or more dimensions, but for the purposes of our model all those aggregated along a "quality" index) but substitutable products for a retailer who faces a long supply lead time and a short selling season. The products are substitutable in the following sense: a customer can pick any one product within the in-stock offerings if

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the product has an appealing quality-price combination. We capture three factors that affect consumer's purchase choice: quality, price, and availability. The retailer has limited control over quality: he makes inventory/procurement decisions by picking from a set of products whose quality levels are exogenous. We explore three main research issues on the dynamic pricing and inventory control decisions for this context:

(1) Given initial inventory levels of substitutable products, what is the optimal dynamic pricing policy for the entire selling season, and what are the key factors that affect the optimal pricing policy?

(2) For what environments do dynamic pricing policies create significant value and thus justify their implementation, besides their potential organizational and computational complexity?

(3) How sensitive is the performance of dynamic pricing to the initial inventory levels, and can we develop a computationally efficient approach to obtain inventory levels that will account for dynamic pricing and substitution and deliver close-to-optimal profit performance?

The answers to those questions shed light on the value and management of dynamic pricing practice in industries that are characterized by short selling seasons within which inventory replenishment is often impossible and noticeable substitution initiated by consumers based on quality-price trade-offs. For example, in the retail sector, seasonal/holiday/promotional sales have relatively short selling periods and inventory is often not replenished because of the long supply lead time from overseas production or a one-time deal with a specific supplier, and products vary in design, quality, and fashion. In the service sector, ticket brokers of theaters and sport events purchase tickets from event organizers at quantity discounts and sell them to the end buyers; tickets expire at the time of the event, and seats are different in viewing quality. In those environments, the substitution effect plays an important role in driving the dynamic pricing adjustments, as it affects not only the overall probability of making a sale but also the sales probability of individual variates within the in-stock variates.

Our model assumes a population of homogeneous (statistically identical and independent) customers making mutually exclusive choices from a set of substitute products. We adopt a discrete choice approach—the multinomial logit (MNL) model (Anderson et al. 1992)—to capture dynamic consumer choice preferences over substitute products as stockouts occur and/or prices are adjusted. The retailer, through the control of inventory and dynamic pricing, affects consumers' choices to his advantage to maximize profit.

As a prelude of results to come, we would like to highlight the following three findings of our research. First, dynamic pricing is of great value to the retailer when inventory is scarce, with the level of inventory scarcity triggering significant benefits of dynamic pricing affected by both procurement costs and differences in quality levels of the substitute products. Second, the optimal price adjustment over time is affected not only by the inventory scarcity at the system (total inventory) level and the individual product level, but also by the quality levels of all the in-stock products. The complex interplay of inventory scarcity and quality difference of substitute products can lead to a wide range of patterns of optimal price trajectories even for a small number of products. Furthermore, the usual intuition of product price changes over time—often derived from single product analysis—as being nonincreasing in inventory and nondecreasing in remaining selling time is easily disproved for the multiproduct case. Third, the performance of dynamic pricing is quite robust to the initial inventory levels when the deviation from the optimal inventory levels is not too great. We develop a computationally efficient approach for the initial inventory decision that delivers close-to-optimal inventory levels for all our testing cases.

1.2. Literature Review

Both dynamic pricing and product substitution have attracted significant attention by researchers from economics, marketing, and operations. From the perspective of operations management, the related literature can be classified into two categories: dynamic pricing of a single product[1] and inventory/assortment/pricing control of substitute products.

[1] Static pricing of a single product is the newsvendor problem with endogenous pricing. Petruzzi and Dada (1999) give a comprehensive review of this problem.

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McGill and van Ryzin (1999), Bitran and Caldentey (2003), and Elmaghraby and Keskinocak (2003) are three important up-to-date reviews on dynamic pricing. The majority of papers reviewed in these studies fall under the first category. Although McGill and van Ryzin cover many issues (forecasting, overbooking, seat inventory control, and pricing) related to revenue management in transportation, Bitran and Caldentey and Elmaghraby and Keskinocak focus on the pricing models for revenue management mainly in the retail environment. Elmaghraby and Keskinocak take inventory into consideration in a dynamic pricing context. We refer interested readers to the above review papers for a complete bibliography.

The operations literature on consumer-initiated substitution can be divided into two streams: static/assortment-driven substitution and dynamic/stockout substitution.[2] Under static substitution, a customer compares products and determines his first choice based on the retailer's assortment, not on the inventory availability: a customer will not initiate any substitution and the sale is lost if his first choice is sold out. Under dynamic or stockout substitution, a customer makes a purchase decision among in-stock offerings.

Papers that assume static substitution often focus on deriving the optimal assortment and inventory policy and/or static pricing policy for a newsvendor setting, where consumer demand distribution for substitute products is determined by the initial assortment and price. Under the static substitution assumption, assortment and pricing decisions determine, in a static manner, the way the potential demand is split among variates in the assortment; inventory availability plays no role in affecting demand splitting. Consequently, when prices are given, optimal inventory levels are straightforward newsvendor results and can be obtained for each variate independently. Therefore, the focus of this stream of research is on optimal assortment and optimal prices. van Ryzin and Mahajan (1999) are the first to adopt the MNL model to determine the normal demand distribution for substitute products and to study the optimal assortment. Aydin and Ryan (2000) use the MNL model to study the joint assortment and pricing problem assuming the firm satisfies demand by operating on a make-to-order basis. Several recent papers, in order to capture intricate consumer shopping behaviors such as searching among multiple retailers (Cachon et al. 2005, 2006) and basket shopping behavior (Cachon and Kök 2007), as well as the retail-level competition, also use various versions of the MNL model and assume make-to-order retailers. Aydin and Porteus (2008) define a broader class of demand models that includes the MNL model as its special case. For this class of demand models, they find the optimal solution to the joint inventory and static pricing policy for a given assortment. Maddah and Bish (2007), building on the van Ryzin and Mahajan (1999) model, provide a joint optimal static pricing and inventory policy and present structural properties of the optimal assortment. Xue and Song (2007) characterize the optimal inventory replenishment and pricing policy in a finite horizon setting where demand of a set of substitute products is random and price sensitive (with the MNL and address models being special cases of the demand model). Because substitution within each period is assortment driven, the joint concavity in inventory and market share can be established.

Dynamic substitution adds complexity to the study of assortment, inventory, and pricing decisions. Because inventory availability does affect the optimal pricing during the selling season, inventory decision and pricing decisions interact in a much more complicated fashion than under the static substitution. Assuming exogenous substitution rates in the case of stockouts, Smith and Agrawal (2000) provide a methodology to solve the joint assortment and inventory problem; Netessine and Rudi (2003) show that the profit function for the problem might have multiple local optima in inventory levels. Under a general discrete choice model, Mahajan and van Ryzin (2001) use a sample path approach to show that for a given assortment, the retailer's profit is not quasiconcave in inventory, and global optimization might be difficult. They provide numerical examples to demonstrate that dynamic substitution can have a significant impact on a given assortment's gross profit. Kök and Fisher (2007) develop a substitution model that takes into

[2] We refer readers interested in supplier-initiated substitution to Bassok et al. (1999), Deniz et al. (2004), and references therein.

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account both static and dynamic substitution. They also propose and test an iterative heuristic for solving the assortment planning problem. Gaur and Honhon (2006) adopt the address model of customer choice to derive the optimal assortment and inventory policy for the static substitution problem; they develop two heuristics for the dynamic substitution case.

The idea of adjusting prices dynamically to affect consumers' substitution behavior over time is appealing but leads to difficult optimization problems. To the best of our knowledge, only a few papers tackle this problem. Gallego and van Ryzin (1997) give the formal formulation of the generalized version of the problems in this category. In their model, n final products are produced from a set of m resources, and the production rates vary across products and resources; the objective is to price the finished products to maximize expected revenue. They point out that in general it is very difficult to find closed form solutions to systems of differential equations representing the Hamilton-Jacobi-Bellman sufficient conditions of their formulated problem (this is the case even for the one-resource-one-product problem, studied in Gallego and van Ryzin 1994). They provide two fixed-pricing heuristics based on the solution to the deterministic version of the stochastic intensity control problem and show that the fixed-pricing heuristics are asymptotically optimal, as the initial inventory and expected sales volume tend to infinity in a constant proportion. Problems in the retail setting, which are the focus of our paper, are a special case of their model in the sense that the resources (supplies) received by the retailer are the final products to be sold by the retailer. This one-to-one correspondence changes the system of equations by loosening its constraints—without, however, removing the difficulty of solving the resulting differential equations. Consequently, even for this one-to-one case, few theoretical results can be found in the literature. Our research represents a step in this direction for the MNL model.

In a recent paper, Maglaras and Meissner (2006) study two revenue-management problems, a dynamic pricing problem and a capacity control problem, in a single-resource-multiproduct setting, a special case of the general model of Gallego and van Ryzin (1997). They show that the two problems can be reduced to a single-product pricing problem that enables a unified treatment. They propose three heuristic algorithms based on the deterministic (fluid) approximation of the new formulation and show their asymptotic optimality. Under a general discrete choice model, Talluri and van Ryzin (2004) formulate the single-leg-multifare-class yield management problem as selecting fare products to offer dynamically and characterize properties of the optimal control policy. Although the single-resource-multiproduct case suits the airline, hotel, and cruise line industries, it is not common in the retail store setting, where products are likely inter-dependent on the demand side but not on the supply side.

Bitran et al. (2006) study dynamic pricing of substitute perishable products under a proposed consumer choice model that captures the effect of budget constraints on consumers' purchasing behavior. They provide optimal pricing decisions for two approximations of the stochastic control problem: the unlimited supply case and the deterministic approximation of the original problem. Zhang and Cooper (2006) propose an approximation of the dynamic pricing of the multisubstitutable-flight problem, which uses a collection of one-flight problems and provides bounds to the optimal revenue. Xu and Hopp (2006) derive the optimal (equilibrium) joint inventory and dynamic pricing decisions for a monopoly firm selling one product (firms in an oligopoly game each selling one product) when the customer arrival process is deterministic or follows a geometric Brownian motion, and they demonstrate the value of pricing flexibility. Our work differs from the above in its chosen consumer choice model (i.e., MNL) and its focus on the effect of the inventory scarcity and quality difference interplay on the optimal price behavior.

Our paper contributes to the literature in four ways. First we present an optimal solution to the dynamic pricing of substitute products with consumers incurring dynamic substitutions throughout the selling season. We show that the optimal dynamic pricing converges to static pricing as inventory levels of all variates approach the number of remaining selling periods (assuming at most one customer arrival within each period). This finding suggests that the value of dynamic pricing is tied to the level of inventory scarcity; i.e., when inventory is scarce, a truly dynamic pricing scheme that takes into account the

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real-time sales and inventory information is likely to significantly outperform a pricing scheme that ignores the real-time information. Thus, our work points out the real need for dynamic pricing scheme to deliver optimal performance when inventory is scarce. Second, our numerical results provide insights on the behavior of the optimal dynamic prices and highlight the complex interplay of inventory scarcity and product quality difference. Quality advantage (disadvantage) strengthens (weakens) the expected effect of the increasing inventory scarcity that increases the optimal prices of individual variates. As a result, the optimal price sample path of a product is far different from that for the single-product case. To the best of our knowledge, this paper is the first to identify factors that affect the behavior of the optimal dynamic pricing in the multiproduct case.

Third we investigate the effectiveness of three alternative pricing schemes that have limited flexibility in price adjustment and suggest environments in which these restricted pricing schemes may work well. We emphasize that the more scarce the initial inventory (which can be a result of inventory decisions for the lower profit potential products) or the wider quality range that a retailer covers in his product offerings, the greater the value of the dynamic pricing policy that takes into account the consumers' dynamic substitution behavior. Finally, as demonstrated by Mahajan and van Ryzin (2001), dynamic substitution creates a difficult problem for the initial inventory decision because the profit function under dynamic pricing is not quasiconcave in the initial inventory levels. In this paper, we develop a computationally efficient approach to inventory decisions. Our numerical testing finds that the profit performance achieved by applying dynamic pricing to the resulting inventory levels is very close to the optimal.

The remainder of the paper is organized as follows: In the next section, we provide a dynamic programming model formulation of the joint optimal starting inventory and dynamic pricing decisions for multiple substitute products. In §3, we present the optimal solution for the pricing problem. In §4, we investigate properties of the optimal dynamic pricing strategy. In §5, we study the value of dynamic pricing by comparing several alternative pricing schemes with the optimal dynamic pricing policy and characterize environments for which dynamic pricing offers significant benefits. In §6, we propose and test a computationally effective approach to the initial inventory decision. We conclude in §7 with a summary of our main results and useful insights from our research. All the proofs are provided in Appendix A, and Appendix B (provided in the online supplement/technical appendix) presents complementary materials to facilitate deeper understanding of results of the paper.

2. Model

2.1. Timing

Time periods are indexed by an integer $t = T_+, T, T-1, \ldots, 0$, with smaller values of t representing later points in time.[3] Specifically, $t = T_+$ represents a time period long before the beginning of the selling season $t = T$, and $t = 0$ represents the end of the selling season. Let $\mathfrak{N}$ be the set of the available product variates in period T_+. At T_+, the retailer places an order that specifies a subset $\mathfrak{n} \subseteq \mathfrak{N}$ with $\mathfrak{n} = \{1, \ldots, n\}$ and the initial inventory $x_T^i \in \mathbb{N}$ for $i = 1, \ldots, n$, where $\mathbb{N}$ is the set of positive integers. We use superscript i to denote variate i and subscript t to denote time periods. We do not differentiate the terms "product" and "variate," but use "variate" whenever referring to a product within a collection of products.

The retailer receives the shipment of the order by the beginning of the selling season $t = T$, when customers start to arrive. We assume customer arrivals follow a Poisson process, with λ being the probability of one arrival within a period. By assuming a sufficiently small time interval as one period, we can ignore the probability of more than one arrival within a period. Thus, the probability of no arrival is $1 - \lambda$. We keep λ as a constant throughout the time horizon; the extension to a time varying λ_t is discussed in §3.2. We assume that the delivery lead time is too long and the selling season is too short to allow replenishment during the selling season. After the order is received

[3] The dynamic pricing problem can be formulated as a continuous-time stochastic intensity control problem. We refer interested readers to Gallego and van Ryzin (1997) for such formulation. Optimal pricing policies developed in this paper for the discrete-time model can be easily translated to the continuous-time setting.

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in period T, in any period $t \le T$ the retailer monitors the inventory level x_t and sets price r_t^i for variate $i \in S(x_t)$, with $S(x_t) \doteq \{i\colon x_t^i > 0\}$. If any variate is sold out before the end of the selling season, it is out of stock for the remaining season because no replenishment opportunity exists. If all variates are sold out in period t, $t > 0$, then the selling is closed at t; otherwise, all unsold variates are salvaged at the end of the selling season.

2.2. Assumptions

If a variate i is out of stock, i.e., $i \notin S(x_t)$, we set $r_t^i = \infty$. Hence, $r_t = \{r_t^1, \ldots, r_t^n\}$ is the price vector in period t. Prices are set using a nonanticipating pricing policy, and the set of allowable prices is the n-dimension nonnegative real numbers. More about nonanticipating pricing policy can be found in Gallego and van Ryzin (1997).

The probability that a customer arriving in period t chooses variate $i \in \mathfrak{n}$ is denoted by $P_t^i(r_t)$. We let $i = 0$ denote the no-purchase choice and $P_t^0(r_t) \ge 0$ denote the no-purchase probability. The condition we impose on the choice probabilities $P_t^i(r_t)$ is that they define a proper probability function. That is, the probabilities satisfy $P_t^i(r_t) \ge 0$ for all $i \in \mathfrak{n}$ and $\sum_{i\in\mathfrak{n}} P_t^i(r_t) + P_t^0(r_t) = 1$. The MNL model satisfies this condition.

Anderson et al. (1992) offer a comprehensive description of consumer discrete choice models including the MNL. Essentially, the MNL model is a special case of the stochastic utility model for a statistically homogeneous population. Within the population, each individual's utility for alternative $i \in \mathfrak{n}$ can be expressed as $U_i = u_i + e_i$, where u_i is a known constant whose value is determined by observable characteristics of the individual and e_i is a random variable whose value is influenced by unobservable characteristics. The probability that an individual selects alternative i is given by $\Pr(U_i = \max_{j\in\mathfrak{n}} U_j)$.

The MNL model assumes that e_is are independently identically distributed following a double exponential distribution with mean 0 and variance $\mu^2\pi^2/6$ ($\mu > 0$). In addition, we assume $u_i = a_i - r_t^i$, where a_i is the quality index of variate i (Anderson et al. 1992, p. 229) and $a_i - r_t^i$ (respectively u_0) represents the known part of the consumer utility associated with variate i (respectively the no-purchase). Thus, for given price vector r_t, we have

$$P_t^i(r_t) = \frac{\exp((a_i - r_t^i)/\mu)}{\sum_{j\in\mathfrak{n}} \exp((a_j - r_t^j)/\mu) + \exp(u_0/\mu)}, \quad (1)$$

$$P_t^0(r_t) = \frac{\exp(u_0/\mu)}{\sum_{j\in\mathfrak{n}} \exp((a_j - r_t^j)/\mu) + \exp(u_0/\mu)}. \quad (2)$$

Because $r_t^i = \infty$ for $i \notin S(x_t)$, (1) implies $P_t^i = 0$ for any $i \notin S(x_t)$. In such a case, r_t^i acts as the *null price*, as defined in Gallego and van Ryzin (1997), which is used to "turn off" the demand for variate i. Thus, the probability of a sale of a unit of variate i in period t is $\lambda P_t^i(r_t)$, and the probability of no sale is $\lambda P_t^0(r_t) + (1-\lambda)$.

As a result of adopting the MNL model in this paper, we restrict our study to environments where homogeneous consumers choose among equally dissimilar products (the removal or addition of one variate affects the purchase probability of all variates in the same proportion, e.g., different color otherwise identical products), which are quite common in retail sectors. The environments where product dissimilarity is asymmetric (the removal or addition of one variate that is more similar to some variates than to others affects the purchase probability of in-stock variates in an asymmetric fashion) can be modeled by a nested MNL model. However, the nested MNL model introduces substantially complex expressions and obscures the fundamental underlying driving forces of the optimal price behavior that we intend to investigate, and thus is outside the scope of the current paper. We refer interested readers to Anderson et al. (1992, p. 46) for more on the nested MNL model and to Kök and Fisher (2007) for consumer choice models that capture other substitution behaviors.

2.3. Dynamic Programming Formulation

For $t = T, T-1, \ldots, 1, 0$, we can write the optimality equation of the dynamic pricing problem as

$$\pi_t(x) = \max_{r_t\in\mathbf{R}_+^n} \Big\{ \lambda \sum_{i\in S(x)} r_t^i P_t^i(r_t) + \lambda \sum_{i\in S(x)} \big[P_t^i(r_t)\pi_{t-1}(x - e^i) + (\lambda P_t^0(r_t) + (1-\lambda))\pi_{t-1}(x) \big] \Big\}, \quad (3)$$

$$\pi_0(x) = 0. \quad (4)$$

Optimality Equation (3) reveals the two effects of dynamic pricing on the retailer's revenue: first, it directly affects the expected current period revenue

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by affecting both the price and sales probability. Second, although not affecting the possible values of the starting inventory in period $t-1$, it affects the *probabilities* that the retailer starts with specific inventory levels in period $t-1$ through its control of the current period sales probabilities. After replacing P_t^0 by $1-\sum_{i\in S(x)} P_t^i$ and rearranging terms in (3), we obtain

$$\pi_t(x) = \max_{r_t\in\mathbf{R}_+^n}\{\xi_t(x, r_t) + \pi_{t-1}(x)\}, \tag{5}$$

where

$$\xi_t(x, r_t) \doteq \sum_{i\in S(x)} \lambda P_t^i(r_t)[r_t^i(x) - \Delta^i\pi_{t-1}(x)],$$

$$\Delta^i\pi_{t-1}(x) \doteq \pi_{t-1}(x) - \pi_{t-1}(x-e^i), \quad i\in S(x).$$

Optimality Equation (5) shows that in period t, with starting inventory x, one only needs to find r_t that maximizes $\xi_t(x, r_t)$. In the expression of $\xi_t(x, r_t)$, r_t^i is the immediate revenue from selling one unit of variate i, and $\Delta^i\pi_{t-1}(x_t)$ is the marginal value of variate i in period $t-1$, i.e., the loss in revenue if the retailer does not have this unit of variate i when entering in period $t-1$, because one unit sale of variate today increases the probability of which variate i is sold out in some future period. Thus, $r_t^i(x) - \Delta^i\pi_{t-1}(x)$ is the marginal revenue from a unit sale of variate i, and $\xi_t(x, r_t)$ is the expected marginal revenue from a unit sale of in-stock variates. We provide the optimal solution to maximizing $\xi_t(x, r_t)$ in the next subsection.

In (4) we assume all salvage values are normalized to zero without loss of generality (the effect of nonzero salvage value will be discussed in §3.2).

Because we consider a short selling season, the holding cost is assumed to be negligible. Also, following the convention in the revenue-management literature, all unsatisfied demand is lost with no penalty cost. Let $c = \{c_1, \ldots, c_n\}$ be the vector of procurement costs. Define the expected profit at time T as

$$V_T(x_T) \doteq \pi_T(x_T) - c'x_T. \tag{6}$$

The initial inventory decision made at time T_+ is then formulated as

$$\max_{x_T\geq 0} V_T(x_T).$$

3. Dynamic Pricing Solution

3.1. Optimal Solution

Hanson and Martin (1996) show by a counterexample that $\xi_t(x, r_t)$ is not quasiconcave in r_t. So the optimal price vector cannot be found directly by concave optimization. Note that (1) and (2) define a function that maps the price space $\mathbb{R}_+^n \cup \{+\infty\}$ to the sales probability space $[0, 1]^n$. Specifically, because $P_t^i/P_t^0 = \exp((a_i - r_t^i - u_0)/\mu)$, we have

$$r_t^i = a_i - u_0 - \mu\ln P_t^i + \mu\ln P_t^0, \quad i = 1, \ldots, n, \tag{7}$$

where $P_t^0 = 1 - \sum_{j=1}^n P_t^j$. Thus, (7) defines an inverse function that maps the sales probability space to the price space $\mathbb{R}_+^n \cup \{+\infty\}$. Consequently, $\xi_t(x, r_t)$ can be rewritten as a function of $P_t \doteq \{P_t^0, P_t^1, \ldots, P_t^n\}$,

$$\xi_t(x, P_t) \doteq \sum_{i\in S(x)} \lambda P_t^i[a_i - u_0 - \mu\ln P_t^i + \mu\ln P_t^0 - \Delta^i\pi_{t-1}(x)]. \tag{8}$$

Although $\Delta^i\pi_{t-1}(x)$ changes as t or x changes, its value is not affected by the firm's action (P_t) prior and up to period t. So, in maximizing $\xi_t(x, P_t)$ by changing P_t, $\Delta^i\pi_{t-1}(x)$ can be treated as a constant, as a_i and u_0. It can be shown that $\xi_t(x, P_t)$ is concave in P_t (see the proof of Theorem 1 in Appendix A). Theorem 1 gives the optimal prices and the maximum expected revenue from period t to 0, $t \leq T$, with any inventory level x_t.

Theorem 1. *An optimal solution to* (5) *is given by*

$$r_t^{i*}(x_t) = \Delta^i\pi_{t-1}(x_t) + m_t(x_t), \quad \forall\, i\in S(x_t), \tag{9}$$

where $m_t(x_t)$ is the unique solution of $m_t(x_t)P_t^0(r_t^*) = \mu$; i.e.,

$$\left(\frac{m_t(x_t)}{\mu} - 1\right)\exp\left(\frac{m_t(x_t) + u_0}{\mu}\right) = \sum_{i\in S(x_t)} \exp\left(\frac{a_i - \Delta^i\pi_{t-1}(x_t)}{\mu}\right). \tag{10}$$

Let $m_s(x_t)$ denote the solution of (10) *with inventory x_t in period $s \leq t$, $s \in [t, 0]$. The maximum expected revenue is*

$$\pi_t(x_t) = \lambda\sum_{s=1}^{t}[m_s(x_t) - \mu]. \tag{11}$$

In (9), the optimal price of variate i should be set so that the marginal revenue $r_t^{i*}(x_t) - \Delta^i\pi_{t-1}(x_t)$ is the same and equals $m_t(x_t)$ across all variates that are still available. For convenience of exposition, we also refer to m_t as the profit margin of selling a unit

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of inventory in period t. In other words, m_t represents the immediate contribution of selling one unit of inventory to the overall revenue, and $\Delta^i \pi_{t-1}$ represents the future value of the variate i inventory to the overall revenue. The equation $m_t P_t^0 = \mu$ specifies the optimal level for m_t. Thus, the higher uncertainty about consumer choice is, the higher the margin the retailer should apply for all variates (straightforward from observing (10)). It is clear from (10) that, *ceteris paribus*, an increase of $\Delta^i \pi_{t-1}$ leads to a decrease of m_t, which reflects a balance between the immediate and future value of a unit of inventory that needs to be reached when determining optimal prices, leading to complex patterns in the optimal price behavior (to be illustrated in §4).

Aydin and Ryan (2000), under the assumption that the retailer can satisfy all customer demand on a make-to-order basis at a constant marginal cost (equivalently, a one period static substitution model), show that all products should be priced to have an identical profit margin, $r^i - c_i = m$. Anderson and de Palma (1992) show results in a similar spirit for a nested logic model. In our context of dynamic substitution with finite initial inventory, the replacement of c_i by $\Delta^i \pi_{t-1}$ in the expression of optimal prices represents the fundamental difference between the dynamic substitution and static substitution—the role of inventory in pricing decisions, which explains the interesting phenomena not existing in any static substitution settings (although c_i is a constant, the value of $\Delta^i \pi_{t-1}$ is affected by inventory levels and the length of the remaining selling season).

It is worth mentioning that the above insights are not readily available from the optimality condition of a continuous time model established by Gallego and van Ryzin (1997) for general demand functions. The MNL assumption allows an explicit expression of the optimal price (9), which leads to the ensuing finding that the value of dynamic pricing is affected by the scarcity level of inventory; this enables us to follow up with a numerical study on the behavior of the optimal price scheme (in §4).

The following corollary, which follows immediately from (9), shows that m_t acts as a lower bound of the optimal prices shared by all variates in stock.

Corollary 1. *For any $t \le T$ and any finite $x_t \in \mathbb{N}^n$, $m_t(x_t) \le r_t^{i*}(x_t)$, $i \in S(x_t)$.*

It is useful to introduce here the notion of inventory scarcity in the context of dynamic substitution. We say that total (individual) inventory is scarce if the total (individual variate) inventory level is lower than the maximum possible total demand (demand to the variate). The larger the shortfall is, the more severe the total (individual) inventory scarcity is. We note that individual inventory scarcity exists even when the total inventory is very high but one or more of the individual variates inventory is less than its maximum possible demand. The reason is that, according to the MNL model, there always exists a positive probability that all arrivals choose one particular variate.

Corollary 2. *If $x_t^i \ge t$ for all $i \in S(x_t)$, then $r_s^{i*} = m$ and $P_s^{i*} = (\mu/m)\exp((a_i - m - u_0)/\mu)$, $P_s^{0*} = \mu/m$, for $s \le t$ and $i \in S(x_t)$, where m is the unique solution of*

$$\left(\frac{m}{\mu} - 1\right)\exp\left(\frac{m + u_0}{\mu}\right) = \sum_{i \in S(x_t)} \exp\left(\frac{a_i}{\mu}\right). \tag{12}$$

Basically, Corollary 2 says that there is no need for dynamic pricing or price differentiation when there is no inventory scarcity. It implies that without inventory scarcity, the marginal value of inventory $\Delta\pi_{t-1}(x)$ is zero, and the myopic policy maximizing $\sum_{i \in S(x_t)} \lambda P_t^i(r_t) r_t^i(x_t)$ is optimal. All products should be priced the same and the induced probabilities of purchasing variates should follow the same order as their quality indices.

When inventory scarcity exists, $\Delta^i \pi_{t-1}(x)$ reflects the individual inventory scarcity of variate i. From the right-hand side of (10) we can clearly see that the value of profit margin m_t depends on the quality level as well as the individual inventory scarcity of every variate in stock, $i \in S(x_t)$. We will demonstrate the impact of $\Delta^i \pi_{t-1}(x_t)$ and m_t on the optimal prices in §4.

3.2. Extensions

The structural results on the optimal pricing can be applied in a straightforward manner to three extensions of the model.[4]

[4] The proofs of those extensions are similar to the original proofs because of the pricing decisions made at time t only affecting the probability of sale, which is easy to account for under the MNL model.

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First, in the case of nonzero salvage value at the end of the selling season, the optimal price in (9) should be adjusted to $r_t^{i*}(x_t) = \Delta^i \pi_{t-1}(x_t) + m_t(x_t) + g_i$, and the optimal expected revenue in (11) should be changed to $\pi_t(x_t) = \lambda \sum_{s=1}^{t} (m_s(x_t) - \mu) + g' x_t$, where $g = (g_i)_{i \in \mathfrak{n}}$ is the salvage value vector.

Second, it can apply to an environment where levels of product quality evolve over time. For example, the quality of perishable goods (such as those in the grocery store) decays as time passes, and products of different ages at the end of the selling season have different salvage values. One way of modeling such changes is to allow quality level a_i to decrease over time in a deterministic fashion. All previous results hold by changing a_i to a_t^i.

Third, it can apply to an environment where the customer arrival rate is nonhomogeneous in time. For example, traffic to stores might vary depending on the days of the week; traffic to websites may vary depending on the time of the day. All previous results hold by changing λ to λ_t.

In the remainder of the paper, we focus on the properties of optimal prices, value of pricing flexibility, and inventory decisions for the original model.

4. Numerical Illustration of Optimal Price Patterns

In this section, we illustrate the complex patterns of the optimal price behavior through numerical examples and discuss the underlying driving forces. Without loss of generality, we consider $\mathfrak{N} = \{1, 2, 3\}$ and order the three variates by their quality indices, i.e., $a_1 \geq a_2 \geq a_3$. The value of the vector a is given below.

Example 1. $a = (11.75, 9, 6.25)$. An example of wide quality range over products.[5]

For simplicity, u_0 is normalized to 0, and μ is normalized to 1. The selling season is partitioned into 100 periods with the expected arrival in each period $\lambda = 0.1$.

To simplify the notation, whenever there is no ambiguity, t is dropped and the subscript is used to denote the variates.

[5] For interested readers we provide an example of narrow quality range over products in online Appendix B.3, which shows similar qualitative results.

It is well known (from Gallego and van Ryzin 1994) for the single variate case that (1) for a fixed inventory level, the optimal price decreases as the length of the remaining selling season decreases, and (2) for a fixed length of selling season, the optimal price decreases as the inventory increases. This result is consistent with our intuition: the less time left or more inventory on hand, the less scarce is the product, and the lower the price should be. Thus, it is clear that the optimal pricing guideline for single product should be: raise the price after a sale of unit; otherwise, decrease the price as time elapses.

One might expect that the pricing guidelines for the single variate case apply to the case of substitute products to some degree. Quite to the contrary, our numerical study shows that for substitute products, not all prices decrease as time elapses; the optimal price change on the sale of one variate is also not obvious. We investigate the changes of r_t^* in the remaining time and the inventory level separately, followed by a sample path analysis.

4.1. Optimal Prices in Time

The main driving forces of the complex behavior of the optimal prices are the *quality difference* among in-stock products, *inventory scarcity*, and the interplay of the two forces.

Figure 1 shows the changes in optimal prices r_t^{i*}, $i = 1, \ldots, 3$, as the length of the remaining selling season t decreases and inventory levels remain constant. Interestingly, even at the same inventory level, the optimal prices for the three variates are not necessarily decreasing as time passes. The price of variate 1 is monotonically decreasing in all three cases, but the prices of variates 2 and 3 in cases (a) and (b) are increasing in the beginning of the selling season.[6] Recall that the optimal price of a variate is the sum of the marginal value of inventory and the profit margin of an immediate sale, $r_t^{i*} = \Delta^i \pi_{t-1} + m_t$. Our observations can be explained by the effect of decreasing inventory scarcity on $\Delta^i \pi_{t-1}$ and m_t (see Figure 1(d)–(f)). Specifically, as the individual inventory scarcity for every variate decreases, the marginal

[6] In the case of only two variates (variates 1 and 2) in stock, the optimal price of variate 2 is not always decreasing in the remaining time either (refer to Figure 5 of Example 1 in Appendix B.3 (online)).

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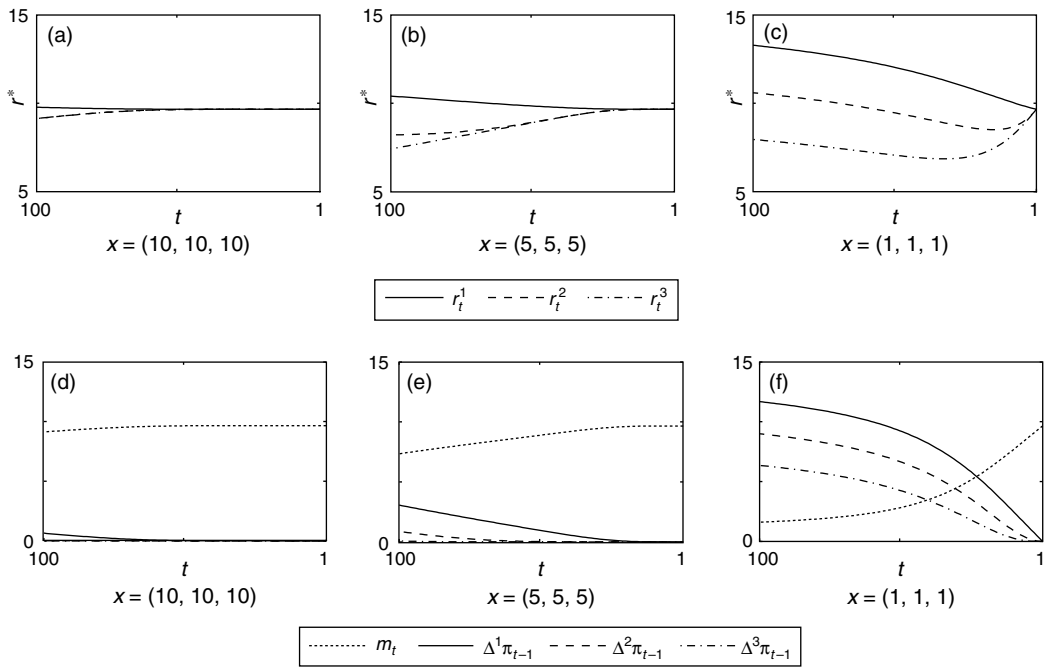
Figure 1 **Optimal Price (a)–(c) and m_t and $\Delta^i \pi_{t-1}$ (d)–(f) in t, Example 1**



value of each variate $\Delta^i \pi_{t-1}$ decreases, but the need to maximize profit in a shorter time frame induces the retailer to increase the profit margin m_t on each unit of inventory. The resulting optimal prices can increase or decrease, depending on the level of inventory scarcity and the product quality level.

In (a) and (b), the retailer starts with moderate inventory levels. The decrease of the marginal value $\Delta^1 \pi_{t-1}$ of variate 1 (the highest quality variate) dominates the increase of profit margin m_t, and variate 1's price decreases, while the prices of variates 2 and 3 (lower quality variate) increase because of a weaker effect of their decreasing marginal value.

In (c), because the inventory is quite scarce for a long period of time ($t > 1$), marginal values $\Delta^i \pi_{t-1}$ are quite high at the beginning of the selling season and then decrease quickly for all three variates; the increase of the margin m_t is relatively slow. Thus, prices of all three variates start high and decrease at the beginning of the selling season.

Corollary 2 states that when inventory is abundant ($x_t^i \geq t$), $r^{i*} = m$, and $P^{i*} = (\mu/m)\exp((a_i - m - u_0)/\mu)$. As time elapses and no sales occur, the inventory scarcity level decreases for every variate. Figure 1 shows the convergence of r_t^{i*} to m.

In summary, the decrease of the inventory scarcity caused by the pass of time increases the optimal profit margin m_t that should be charged for each variate but decreases the marginal value of inventory $\Delta^i \pi_{t-1}$, with the slope of decrease determined by the quality difference among variates. The decrease of $\Delta^i \pi_{t-1}$ is more (less) significant for high (low) quality variate when compared to the increase of m_t, leading to the decrease (increase) of its optimal price.[7] But severe inventory scarcity strengthens the impact of $\Delta^i \pi_{t-1}$ on the optimal prices, and the optimal price can even decrease for the lowest quality variate.

4.2. Optimal Prices in Inventory

We now examine the relationship between the optimal prices and the inventory level. Figure 2 shows the change in optimal prices when the inventory level of variate 1 decreases and the inventory levels of other variates and the length of the selling season are fixed.

For the sake of convenience, we always use i to denote the variate whose inventory changes and j to

[7] For the single product case, in the absence of the effect of quality difference, the decrease of $\Delta^i \pi_{t-1}$ always dominates. The price decreases as time elapses.

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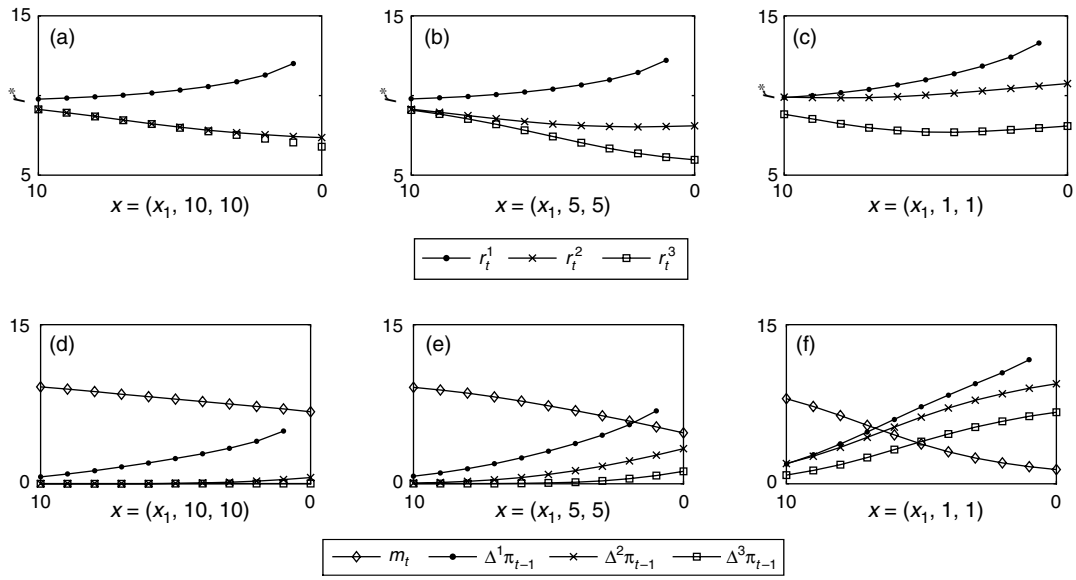
Figure 2 **Optimal Price (a)–(c) and m_t and $\Delta^i \pi_{t-1}$ (d)–(f) in x_1 for $t = 100$, Example 1**



refer to either of the other two variates whose inventory level is fixed. We use the term *self-effect* to refer to the impact of x_i on r_i^* and the term *cross effect* to refer to the impact of x_i on r_j^*, $j \neq i$, $i = 1, 2, 3$.

For substitute variates, the usual intuition is that r_i^* should increase and r_j^* should decrease as x_i decreases. Figure 2 shows that such intuition may not apply here. Although r_i^* does increase as x_i decreases, r_j^* can decrease (e.g., (a) and (b)) or increase (e.g., (c)).

We argue here again that the joint effect of quality difference and inventory scarcity affects the change of optimal prices. On the one hand, the increase of variate i's inventory scarcity leads to the increase of its marginal value $\Delta^i \pi_{t-1}$. The increase of the total inventory scarcity increases the marginal value of the other variates in stock; i.e., $\Delta^j \pi_{t-1}$ also increases. Variate i has the greatest increase in marginal value, and variate j has less marginal value depending on its quality level. On the other hand, the increase of the total inventory scarcity decreases the profit margin m_t that the retailer would apply on each unit of inventory in stock. Therefore, the opposite moving directions of $\Delta \pi_{t-1}$ and m_t in response to the decrease of variate i inventory create the complexity in the overall optimal price movement.

We look at the self-effect first. The large increase in $\Delta^i \pi_{t-1}$ outweighs the decrease in m_t. Therefore, variate i's price increases as its inventory level decreases.

The cross effect of variate i on any other variate j in stock is not as clear as for the self-effect. As the increase of $\Delta^j \pi_{t-1}$ depends on its quality level and the severity of its individual inventory scarcity, r_t^j can increase or decrease.

Take the inventory change of variate 1, the highest quality product, for example. Figure 2 shows that as the inventory of variate 1 decreases, the increase of the marginal value ($\Delta^2 \pi_{t-1}$ and $\Delta^3 \pi_{t-1}$) of the less attractive variates 2 and 3 is dominated by the decrease of the profit margin m_t, and thus the optimal prices of variates 2 and 3 decrease. The only exception is case (c), where the severity of total inventory scarcity leads to strong enough increases of $\Delta^2 \pi_{t-1}$ and $\Delta^3 \pi_{t-1}$ to overcome the decrease in m_t. This intuition carries through to cases where inventory changes happen to variate 2 or variate 3 (see Figure 4 in Appendix B.3 (online)).

In summary, as variate i's inventory depletes, the increase of the total inventory scarcity leads to the increase of variate j's marginal value $\Delta^j \pi_{t-1}$ and the decrease of the profit margin of each variate m_t. The quality difference among variates can affect the magnitude and sign of the combined effect on optimal prices. In particular, the quality advantage (disadvantage) of variate j can strengthen (weaken) the increase of $\Delta^j \pi_{t-1}$, leading to the increase (decrease) in variate j's price. However, severe inventory scarcity can

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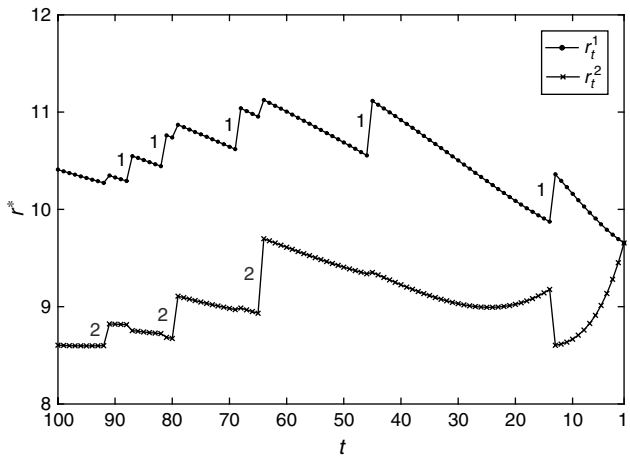
Figure 3 Sample Path of the Optimal Prices, Variates 1 and 2 of Example 1



compensate for the quality disadvantage of variate j by strengthening the increase of $\Delta^j \pi_{t-1}$, leading to the price increase even for the lowest quality variates.

4.3. Sample Path Analysis

We now present a sample path to show how the optimal prices change when both the inventory and the remaining time change. Figure 3 is a sample path of the optimal prices for Example 1. To keep the graph clear and easy to understand, we start from the inventory level $x_{100} = (6, 4, 0)$ at time $t = 100$.

There are eleven total arrivals in this sample path, among which five purchase variate 1, three purchase variate 2, and three purchase nothing. The leftover inventory is $(1, 1, 0)$. The purchases of variate 1 occur at time $t = 88, 82, 69, 46, 14$. The price of variate 1 jumps up at each of the purchase points. Although r_t^{2*} changes every time r_t^{1*} jumps up, the change may be a jump ($t = 69, 46$) or a drop ($t = 88, 82, 14$). Similarly, the purchases of variate 2 occur at time $t = 92, 80, 65$, where the price of variate 2 always jumps up, and the price of variate 1 always jumps up, too. Also, the price of variate 2 increases toward the end of the selling season even when no sales occur.

5. Value of Pricing Flexibility

The goals of this section are to understand the value of the full-scale dynamic pricing relative to the static or partial-dynamic pricing schemes and to answer the questions in what environments dynamic pricing can provide significant improvement to profit and in what environments other pricing schemes are reasonably effective.

We compare the profit performance of the dynamic pricing studied in the previous sections with three pricing schemes that are of limited price adjustment flexibility. The three pricing schemes are: static pricing, unified dynamic pricing, and mixed dynamic pricing. We will first present the optimal price policy for each pricing scheme and then compare profit performance numerically for the case of two variates.

5.1. Three Pricing Schemes

5.1.1. Static Pricing. Under static pricing, the firm applies fixed prices throughout the selling season. There are different ways to determine the fixed prices; we will consider two of them. The first approach assumes static substitution (hereafter, referred to as static-static approach, where the second static means pricing policy is obtained by assuming static substitution); i.e., a customer does not purchase if his preferred variate in the initial assortment is out of stock. The demand for variate i is assumed to follow a normal distribution with mean $\lambda T P_i$ and standard deviation $\sqrt{\lambda T P_i}$, where P_i is defined as in (1), λ is an exogenous Poisson arrival rate of customers, and T is the length of the selling season. This normal demand distribution is an approximation of demand generated by a Poisson process with rate λ and time horizon T. van Ryzin and Mahajan (1999) use this demand model to study optimal assortment, and Maddah and Bish (2007) study the joint pricing, assortment, and inventory decisions. We use the optimal prices and initial inventory levels from the algorithm provided by Maddah and Bish (2007) but calculate the profit by applying those prices and inventory levels to the dynamic substitution environment, where a customer makes a purchase decision within the in-stock products.

The attractiveness of this approach is that the pricing decision does not require real-time inventory information and given prices, inventory decisions are straightforward newsvendor results. For all other pricing schemes discussed in this section, we have to obtain the optimal inventory levels through exhaustive search.

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The second approach derives the optimal fixed prices by taking full account of dynamic substitution in the selling season. The detailed derivation of fixed prices under this approach is provided in Appendix B.1 (online).

5.1.2. Unified Dynamic Pricing. Unified dynamic pricing applies an identical price to all variates, and this price can be changed dynamically. Liu and Milner (2006) provide excellent examples for such a pricing strategy in apparel, sporting goods, and airline industries and provide heuristic algorithms for the case of static substitution. For a dynamic substitution environment, the optimal dynamic price that applies to all variates at the same time is provided by Proposition 1.

Proposition 1. *For any remaining time $t = T, \ldots, 0$ and inventory level x_t, the optimal unified dynamic price is given by*

$$r_t^*(x_t) = \Delta\pi_{t-1}^w(x_t) + m_t(x_t), \tag{13}$$

where

$$\Delta^w \pi_{t-1}(x_t) = \frac{\sum_{i\in S(x_t)} \exp(a_i/\mu)\Delta^i \pi_{t-1}(x_t)}{\sum_{i\in S(x_t)} \exp(a_i/\mu)}$$

and $m_t(x_t)$ is the unique solution of $m_t(x_t)P_t^0(r_t^*) = \mu$, i.e.,

$$\left(\frac{m_t(x_t)}{\mu} - 1\right)\exp\left(\frac{m_t(x_t)+u_0}{\mu}\right) = \exp\left(\frac{a_w - \Delta^w \pi_{t-1}(x_t)}{\mu}\right),$$

where $a_w = \mu \ln(\sum_{i\in S(x_t)} \exp(a_i/\mu))$. The maximum expected revenue is

$$\pi_t(x_t) = \lambda \sum_{s=1}^{t} [m_s(x_t) - \mu]. \tag{14}$$

In Proposition 1, m_t still represents the immediate contribution of selling a unit of inventory to the overall revenue, but $\Delta^w \pi_{t-1}(x_t)$ is now the weighted future value of a unit inventory of all variates $i \in S(x_t)$ to the overall revenue.

5.1.3. Mixed Dynamic Pricing. Mixed dynamic pricing is a combination of static pricing and dynamic pricing. Suppose a retailer with a wide product offering is considering running both catalogue and online sales. The retailer can select products to advertise in the catalogue to attract consumers to its website, where full lines of products are offered for sale. Products sold exclusively via the Internet are dynamically priced, and the catalogue sales are at the advertised prices.

Without loss of generality, let $\mathfrak{n}_1 = \{1, \ldots, n_1\}$ and $\mathfrak{n}_2 = \mathfrak{n} \backslash \mathfrak{n}_1$ denote the set of the dynamic-pricing and fixed-pricing products in the initial assortment, respectively. Also, let $S_1(x_t)$ and $S_2(x_t)$ denote the set of the in-stock dynamic-pricing and fixed-pricing variates in period t with inventory of x_t, respectively. Then $\mathfrak{n} = \mathfrak{n}_1 \cup \mathfrak{n}_2$ and $S(x_t) = S_1(x_t) \cup S_2(x_t)$. Define r^j as the fixed price of any variate $j \in \mathfrak{n}_2$. Proposition 2 characterizes the optimal dynamic prices for variates in $S_1(x_t)$.

Proposition 2. *Suppose the prices of the variates in $\mathfrak{n}_2$ are fixed. For any remaining time $t = T, \ldots, 0$ and inventory level x_t, the optimal dynamic prices of the variates in $S_1(x_t)$ are given by*

$$r_t^{i*}(x_t) = \Delta^i \pi_{t-1}(x_t) + m_t(x_t), \quad \forall\, i \in S_1(x_t),$$

where $m_t(x_t)$ is the unique solution of

$$\sum_{j\in S_2^+(x_t)} \left(\left(\frac{m_t(x_t) - v_j}{\mu} - 1\right)\exp\left(\frac{m_t(x_t) + u_j}{\mu}\right)\right) = \sum_{i\in S_1(x_t)} \exp\left(\frac{a_i - \Delta^i \pi_{t-1}(x_t)}{\mu}\right), \tag{15}$$

where $S_2^+(x_t) = S_2(x_t) \cup \{0\}$, $v_0 = 0$, $v_j = r^j - \Delta^j \pi_{t-1}(x_t)$, and $u_j = a_j - r^j$ for all $j \in S_2(x_t)$.

The maximum expected revenue is

$$\pi_t(x_t) = \lambda \sum_{s=1}^{t} [m_s(x_t) - \mu].$$

In Proposition 2, m_t also represents the immediate contribution of selling a unit of inventory to the overall revenue, and $\Delta^i \pi_{t-1}(x_t)$ represents the future value of variate i inventory to the overall revenue for all $i \in S_1(x_t)$. Intuitively, in determining m_t for the variates in $S_1(x_t)$, besides the quality and inventory of the variates in $S_1(x_t)$ itself, we have to take into account both the utility of the outside choice and the quality and inventory of the variates in $S_2(x_t)$. In fact, from (15) we know that the variates in set $S_2(x_t)$ play the same role as the outside choice does in the full-scale dynamic pricing model. Clearly, if the fixed-pricing set is empty, (15) is identical to (10). We also note that $m_t P_t^{0*} = \mu$ does not hold in the mixed pricing model. Now m_t satisfies

$$m_t P_t^{0*} + \sum_{j\in S_2(x_t)} P_t^{j*}[m_t - r^j + \Delta^j \pi_{t-1}(x_t)] = \mu.$$

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The mixed dynamic pricing scheme introduces a new challenge of partitioning products into two sets. It is not intuitively obvious what the optimal partition should be. We provide some insights in the following subsection.

5.2. Performance Comparison

To compare the profit performance of three pricing schemes with the full-scale dynamic pricing, we choose two sets of quality indices, $a = (11.75, 9)$ and $a^\dagger = (10.75, 10)$, which represent wide and narrow quality difference, respectively, and two sets of procurement costs, $c = (9, 7)$ and $c^\dagger = (2, 2)$, which represent high and low unit procurement costs respectively. The combinations give us four examples, II–V, as illustrated in Table 1.

Table 1 Parameters of Examples II–V

Example	$c = (9, 7)$	$c^\dagger = (2, 2)$
$a = (11.75, 9)$	II: a, c	III: $a, c^\dagger$
$a^\dagger = (10.75, 10)$	IV: $a^\dagger, c$	V: $a^\dagger, c^\dagger$

Table 2 presents a comparison of the optimal initial inventory level x^*, the maximum expected profit V^* for all pricing schemes, and the corresponding profit loss with respect to the optimal profit from dynamic pricing $\eta = (V_d^*(x_d^*) - V_k^*(x_k^*))/V_d^*(x_d^*)$, where d is an abbreviation of dynamic and $k = s, u, m$ are abbreviations of static, unified, and mixed, respectively.[8] In the mixed pricing model, if variate i's price is fixed, we find its optimal value r_i^* by an exhaustive search over the domain, so that the value of V_m^* is the maximum that a mixed pricing policy can achieve by fixing r_i, $i = 1, 2$, and the comparison is impartial.

We have four observations: First, all three alternative pricing schemes perform well in the column of low unit procurement cost $c^\dagger = (2, 2)$. This is because the relatively high profit potential $(a - c)$ for both products gives incentives for the retailer to stock up, resulting in low inventory scarcity, and the advantage of the full-scale dynamic pricing over static or partial-dynamic pricing is quite small. In contrast, the value of dynamic pricing is significant in the column of high unit procurement cost $c = (9, 7)$, when inventory scarcity is moderate or high, in which cases the performance of static pricing declines quickly, especially the static-static approach.

Second, the profit performance of unified dynamic pricing is closest to that of the full-scale dynamic pricing. One should recognize that price adjustments under unified pricing can only affect the overall sales probability of the in-stock products, not the relative sales probability of a particular variate within the in-stock offerings, because the relative sales probability between a pair of variates $P^i/P^j = \exp((a_i - a_j - r^i + r^j)/\mu) = \exp((a_i - a_j)/\mu)$ remains constant until one variate is sold out. The impact of such inflexibility is profound when products vary greatly in their quality indices. For such cases, higher-quality products are likely to be sold out sooner than lower-quality products and the assortment's overall attractiveness to the consumer decreases quickly with such stockouts. In the column of $c = (9, 7)$, the performance of unified pricing declines as the quality difference increases. Thus, this pricing scheme is most effective when products' quality levels are quite close. Interpreting this result for apparel retailers, who are restricted to applying the same price to clothes of different colors, if some colors are much more popular than other colors, then changing to the full-scale dynamic pricing can improve the retailer's profit significantly. In the case of ticket brokers selling tickets to end buyers, dynamically pricing seats in the same

Table 2 Performance Comparison of Pricing Schemes

	Example II			Example III		
Policy	x^*	V^*	η (%)	x^*	V^*	η (%)
Dynamic	(3, 2)	8.96		(10, 1)	60.74	
Static-dynamic	(3, 2)	7.49	16.40	(10, 1)	59.90	1.38
Static-static	(3, 1)	7.02	21.65	(11, 1)	59.75	1.63
Unified	(4, 1)	8.32	7.14	(10, 1)	60.67	0.12
Mixed (fix r_2)	(4, 1)	8.25	7.90	(11, 0)	60.60	0.23
Mixed (fix r_1)	(3, 2)	7.65	14.62	(11, 0)	59.78	1.58
	Example IV			Example V		
Dynamic	(1, 4)	9.85		(7, 4)	54.93	
Static-dynamic	(1, 4)	7.99	18.88	(7, 4)	54.64	0.53
Static-static	(1, 4)	7.86	20.20	(8, 4)	54.27	1.20
Unified	(0, 5)	9.55	3.05	(7, 4)	54.77	0.29
Mixed (fix r_2)	(2, 3)	7.70	21.83	(8, 3)	54.53	0.73
Mixed (fix r_1)	(0, 5)	9.55	3.05	(7, 4)	54.30	1.15

[8] Comparing their performances at their corresponding optimal initial inventory level yields the smallest possible performance gaps.

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(or neighboring) section(s) but of very different viewing qualities is a smart practice.

Third, the performance of mixed dynamic pricing varies considerably depending on the choice of products for fixed pricing. Mixed pricing performs almost as well as unified pricing when choosing products with low profit potential (i.e., with low $a-c$) to price them at fixed prices and high profit potential products to price dynamically. Our previous finding that dynamic pricing is most effective in improving *revenue* when inventory level is low (i.e., high inventory scarcity) does not contradict this result. From a *profit* maximization perspective, high profit-potential products offer wider ranges for price adjustment, which in turn can influence consumer choice more effectively; thus, dynamic pricing applies the best to high profit-potential products. This observation has useful implications for the previous example of the retailer selling through dual channels of catalogue and online: listing low profit-potential products in the catalogue, where prices are specified and customers expect them to be fixed, is a good strategy from both a marketing perspective (attracting customers through "good bargains" in the catalogue) and an operations perspective (an effective dynamic pricing strategy). The more exotic products with added features and luxury specifications will be offered online via dynamic prices.

Finally, in all four examples, the optimal initial inventory levels of the alternative pricing schemes are all quite close to the optimal inventory levels of the full-scale dynamic pricing; the pricing scheme that takes into account the dynamic substitution in the selling season provides closer-to-optimal inventory levels than the static pricing that assumes the static substitution. This observation suggests that the initial inventory levels are relatively insensitive to the pricing scheme employed in the season.

In the next section, we confirm this observation for a larger set of examples and explore its implication for the initial inventory decision.

6. Inventory Decisions

The obvious linkage between inventory decisions and the use of dynamic pricing prompts the following questions. First, if a retailer employs the dynamic pricing scheme, how sensitive is his profit performance to the initial inventory levels? In other words, can dynamic pricing "overcome" the inefficiency created by a "wrong" initial inventory decision? Second, it is well known that profit function is not quasiconcave in the inventory when consumer substitution is stockout driven (Mahajan and van Ryzin 2001). Can we develop a computationally efficient approach to produce initial inventory levels for dynamic pricing to deliver reasonably good profit performance?

To answer the first question, we vary the allocation of the initial total inventory level that is optimal for the dynamic pricing scheme and compare the profit performance of applying dynamic pricing. Table 3 summarizes the relative loss by using the wrong inventory allocations for Examples II–V in §5. Three observations are in order. First, in general the profit performance of dynamic pricing is not robust to the inventory allocation decisions, with the worst allocation of inventory resulting in 68% profit loss. Second, comparing cases of high inventory scarcity (Ex. II and IV) and cases of low inventory scarcity (Ex. III and V), the profit performance is relatively more robust when total inventory scarcity is low. Third, in all four examples, profit performance is quite robust in the neighborhood of the optimal inventory allocation, especially when the "wrong" inventory allocation is in the "correct" direction (if $x_1^* > x_2^*$ in (x_1^*, x_2^*), the profit of (x_1^*+1, x_2^*-1) is higher than the profit of (x_1^*-1, x_2^*+1)).

Table 3 **Relative Profit Loss Due to Wrong Inventory Allocation (in %), Examples II–V**

x	(5, 0)	(4, 1)	(3, 2)	(2, 3)	(1, 4)	(0, 5)						
Ex. II	12.46	1.14	0	6.63	21.09	45.88						
Ex. IV	68.57	39.36	19.32	6.19	0	3.04						
x	(11, 0)	(10, 1)	(9, 2)	(8, 3)	(7, 4)	(6, 5)	(5, 6)	(4, 7)	(3, 8)	(2, 9)	(1, 10)	(0,11)
Ex. III	0.24	0	0.73	2.11	4.08	6.68	9.91	13.79	18.38	23.73	30.02	37.68
Ex. V	4.89	2.14	0.71	0.07	0	0.42	1.31	2.73	4.73	7.45	11.09	16.19

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Observations from Table 3 suggest that one can achieve close-to-optimal profit performance if the total inventory level as well as the inventory allocation are reasonably close to the optimal. The challenge of the multivariate dynamic substitution problem is that the profit function is not quasiconcave in inventory, and the optimal inventory levels has to be obtained through exhaustive search. In contrast, the profit function of the single variate dynamic pricing problem is concave in inventory (Gallego and van Ryzin 1994, Theorem 1), and thus the optimal inventory level can be obtained quickly. One way to achieve computational efficiency is to make use of this property of the single variate problem. We now propose a two-step approach that produces inventory levels for the n-variate problem with quality indices $\{a_1, a_2, \ldots, a_n\}$ and cost vector $\{c_1, c_2, \ldots, c_n\}$.

Synthesization-Separation (SS) Approach

Step 1 (Synthesization). (1) Construct a pseudovariate with quality index and unit procurement cost defined as below:

$$a^* = \mu \ln\left(\sum_{i=1}^{n} \exp(a_i/\mu)\right),$$

$$c^* = \sum_{i=1}^{n} \kappa_i c_i,$$

where

$$\kappa_i = \frac{\exp((a_i - c_i)/\mu)}{\sum_{j=1}^{n} \exp((a_j - c_j)/\mu)}, \quad i = 1, \ldots, n.$$

(2) Obtain the optimal inventory level $\bar{x}$ for the single variate problem obtained from Step 1, assuming dynamic pricing is employed in the selling season.

Step 2 (Separation). Obtain inventory levels for all variates $x^*_{ss} = (x^*_{ssi})_{i=1}^n$ by

$$x^*_{ssi} = Round(\kappa_i \bar{x}),$$

where function $Round(x)$ rounds x to the closest integer.

The synthesization step of the SS approach is based on the conceptual similarity between dynamic pricing of substitute products and inventory pooling. That is, adjusting prices based on the inventory conditions so as to affect consumer substitution probability dynamically is analogous to pooling inventories of all variates to satisfy consumer demand. Thus, the quality index a^* of the pseudovariate should capture as much as possible the overall attractiveness of the available variates. The setting a^* in Step 1 yields a no-purchase probability of the pseudovariate that is identical to the no-purchase probability of the assortment when the same price is applied to all variates and all variates are in stock. The unit purchase cost of the pseudovariate is a weighted sum of the unit costs of all variates with weights κ_is reflecting variates' relative attractiveness of potential profits.

The separation step is motivated by the observation from Table 2 that variates of higher profit potential value $a - c$ have higher initial inventory levels. The proportion of the individual inventory to the total inventory should reflect the relative (profitability) attractiveness of the individual variate, which is better described by exponential weights that are intrinsic to the MNL choice model than by other weights such as linear.

We define the performance deviation of the SS approach from the optimal by $\delta \doteq (V^*(x^*) - V^*(x^*_{ss}))/V^*(x^*)$, where x^* is the true optimal inventory and $V^*(x)$ is the expected profit of applying dynamic pricing on initial inventory x. Table 4 (Table 5) shows the optimal initial inventory level x^* (at the top, obtained from exhaustive search) and x^*_{ss} for 57 two-variate cases (45 three-variate cases). The SS approach delivers the optimal inventory in 46 of 57 cases (32 of 45 cases). For the 11 (13) cases in which the inventory levels differ, the difference in total inventory is at most 1, and the allocation of inventory is very close to the optimal allocation, skewed in the desirable direction; 9 of the 11 cases (9 of 13 cases) yield less than 1% profit reduction, and the worst case has 3.61% (5.06%) in profit loss.

The robustness of the inventory performance of the SS approach confirms our initial intuition that dynamic pricing of substitute products achieves an effect similar to inventory pooling; therefore, the SS approach, which aggregates all variates into one pseudovariate, effectively pooling all inventories, delivers good inventory results. Computationally, solving the optimal initial inventory level for a single variate problem is much less time demanding than for a multivariate problem. The weakness of the SS approach shows when products have low profit potentials (e.g., $c_i = 7/9a_i$). In those cases, inventory stockouts can happen quite often, and the overall attractiveness of

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Table 4 Inventory and Profit Performance of the SS Approach: Two Variates

(a_1, a_2)	(9, 9)	(10, 9)	(11, 9)	(12, 9)	(13, 9)	(14, 9)	(15, 9)	(16, 9)	(17, 9)	(18, 9)
$c_i = 2$	(5, 5)	(7, 3)	(10, 1)	(10, 1)	(11, 0)	(11, 0)	(12, 0)	(12, 0)	(12, 0)	(12, 0)
$c_i = 4.5$	(4, 3)	(6, 2)	(7, 1)	(8, 0)	(9, 0)	(9, 0)	(10, 0)	(10, 0)	(10, 0)	(10, 0)
$c_i = 7$	(2, 2)	(4, 1)	(6, 0) (5, 1)[1]	(6, 0)	(7, 0)	(8, 0)	(8, 0)	(8, 0)	(9, 0)	(9, 0)
$c_i = \frac{2}{9}a_i$	—	(7, 3)	(8, 2)	(9, 1)	(10, 1) (10, 0)[2]	(10, 1) (10, 0)[3]	(10, 1) (11, 0)[4]	(11, 0)	(11, 0)	(11, 0)
$c_i = \frac{1}{2}a_i$	—	(4, 3)	(5, 2)	(6, 2) (6, 1)[5]	(7, 1)	(7, 1)	(7, 1) (8, 0)[6]	(7, 1) (8, 0)[7]	(8, 0)	(8, 0)
$c_i = \frac{7}{9}a_i$	—	(2, 2) (3, 2)[8]	(3, 2)	(3, 2)	(3, 2) (4, 1)[9]	(4, 1)	(4, 1)	(4, 1)	(4, 1) (5, 1)[10]	(4, 1) (5, 1)[11]

Notes. δ_i represents the relative profit loss due to the suboptimal initial inventory level generated by the SS approach, where subscript i corresponds to the superscripted footnote in the table. $\delta_1 = 0.56\%$, $\delta_2 = 0.29\%$, $\delta_3 = 0.15\%$, $\delta_4 = 0.08\%$, $\delta_5 = 0.30\%$, $\delta_6 = 0.63\%$, $\delta_7 = 0.02\%$, $\delta_8 = 0.39\%$, $\delta_9 = 0.39\%$, $\delta_{10} = 3.61\%$, $\delta_{11} = 2.27\%$.

the products in stock will be lower than the attractiveness of the pseudovariate in the SS approach. Thus, inaccuracy created by synthesization tends to be large.

An alternative is to apply the approach of Maddah and Bish (2007), which is also computationally efficient but is designed for the static substitution environment, to solve for the initial inventory levels. Our numerical study of the 57 two-variate cases in Table 4 (see Appendix B.3 (online)) shows that the resulting inventory and profit performance are both dominated by the SS approach: 28 of the 57 cases offer inventory levels different from the optimal, 11 of the 28 cases yield more than 1% profit reduction, and the worst case has 11.23% profit loss. The comparison again suggests that for a dynamic substitution environment, inventory decision making that reasonably reflects the nature of the substitution will produce much more robust inventory and profit performance than the approach that simplifies the dynamic substitution to a static one.

7. Conclusions

The real-time management of agile supply chains for better matching of supply and demand over very short time intervals necessitates the use of active demand management practices that fully reflect supply conditions. Dynamic pricing policies of substitute goods over a selling season in response to realized sales conditions over the season are effective examples of such real-time active demand management practices.

Table 5 Inventory and Profit Performance of the SS Approach: Three Variates

(a_1, a_2, a_3)	(9, 9, 9)	(9, 9, 6)	(12, 9, 9)	(12, 9, 6)	(12, 12, 9)	(15, 9, 9)	(15, 12, 9)	(15, 12, 6)
$c_i = 2$	(4, 3, 3)	(5, 5, 0)	(10, 1, 0)	(10, 1, 0)	(6, 5, 0)	(12, 0, 0)	(11, 1, 0)	(11, 1, 0)
$c_i = 4.5$	(3, 3, 2) (3, 2, 2)[1]	(4, 3, 0)	(8, 0, 0) (8, 1, 0)[2]	(8, 0, 0)	(5, 4, 0)	(10, 0, 0)	(10, 0, 0)	(10, 0, 0)
$c_i = 7$	(2, 2, 1)	(2, 2, 0)	(6, 0, 0)	(6, 0, 0)	(4, 3, 0)	(8, 0, 0)	(8, 0, 0)	(8, 0, 0)
$c_i = \frac{2}{9}a_i$	—	(5, 4, 1) (5, 5, 0)[3]	(8, 1, 1)	(9, 1, 0)	(5, 5, 1) (5, 5, 0)[4]	(10, 1, 0) (11, 0, 0)[5]	(9, 2, 0) (10, 1, 0)[6]	(9, 2, 0) (10, 1, 0)[7]
$c_i = \frac{1}{2}a_i$	—	(3, 3, 1)	(5, 2, 1) (6, 1, 1)[8]	(6, 1, 1) (6, 2, 0)[9]	(4, 3, 1)	(7, 1, 0)	(6, 2, 0)	(6, 2, 0)
$c_i = \frac{7}{9}a_i$	—	(2, 2, 1)	(3, 1, 1)	(3, 1, 1) (3, 2, 1)[10]	(2, 2, 1)	(3, 1, 1) (4, 1, 1)[11]	(3, 1, 1) (3, 2, 1)[12]	(3, 2, 0) (4, 2, 0)[13]

Notes. δ_i represents the relative profit loss due to the suboptimal initial inventory level generated by the SS approach, where subscript i corresponds to the superscripted footnote in the table. $\delta_1 = 0.03\%$, $\delta_2 = 0.95\%$, $\delta_3 = 0.21\%$, $\delta_4 = 0.43\%$, $\delta_5 = 0.08\%$, $\delta_6 = 0.05\%$, $\delta_7 = 0.05\%$, $\delta_8 = 0.003\%$, $\delta_9 = 0.08\%$, $\delta_{10} = 5.06\%$, $\delta_{11} = 3.24\%$, $\delta_{12} = 2.38\%$, $\delta_{13} = 5.04\%$.

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Our paper obtains, within a stylized modeling framework, the optimal dynamic pricing of substitute products. This pricing policy results from a stochastic dynamic programming formulation of the problem. The optimality condition reveals that the optimal pricing should balance the profit returns from all variates while taking into account the inventory scarcity across all variates. We have shown that it is optimal to adjust prices dynamically only when inventory is scarce. Our numerical studies have demonstrated that the optimal price changes are affected not only by the inventory scarcity at the system (total inventory) level and the individual variate level but also by the quality of all in-stock variates. Unfortunately, the complex interplay between inventory scarcity and quality differences does not allow for provable general properties for the dynamically optimal pricing polices. However, it does allow the disproving of faulty intuition originating from single variate settings on monotonicity properties for prices as functions of remaining inventory and/or selling time.

We have presented and analyzed three alternative pricing schemes (static, unified dynamic, and mixed dynamic pricing schemes) that have limited flexibility in adjusting prices in a fully dynamic fashion because of lack of an information system or conventions in practice. When dynamic pricing is contrasted with the performance of any of the restricted pricing schemes, and especially the static one, it offers great value for environments of inventory scarcity. Such results reinforce the value of real-time pricing of inventories fully reflecting stocking levels and quality differences of all products still in stock.

Finally, our numerical study finds that the profit performance of dynamic pricing is quite robust to the initial inventory levels when they are reasonably close to the optimal inventory levels, and while inventory decisions are robust in the pricing schemes employed in selling season. Based on this observation, we proposed and tested a computationally efficient approach that can generate initial inventory levels that enable dynamic pricing to deliver close-to-optimal profit performance. Although we have restricted our efforts in this paper on an MNL customer choice model, future research should study the effects of different consumer choice models on the performance of dynamic pricing policies and inventory levels, and should take into consideration consumers' strategic response to retailers' dynamic pricing practice (see Huchzermeier et al. 2002, Aviv and Pazgal 2008, Su 2007, and Elmaghraby et al. 2008).

Electronic Companion

An electronic companion to this paper is available on the *Manufacturing & Service Operations Management* website (http://msom.pubs.informs.org/ecompanion.html).

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Appendix A. Proofs of Properties and Theorems

Remark. To simplify notation, throughout this section we drop x_t in $r_t(x_t)$ and $m_t(x_t)$ and r_t in $P_t(r_t)$ whenever there is no ambiguity. Also, we use S_t for $S(x_t)$, and S_{ti} for $S_i(x_t)$, $i = 1, 2$, for simplicity.

Proof of Theorem 1. Without loss of generality, we assume $S_t = S(x_t) = \{1, 2, \ldots, n\}$. The optimal price vector r_t^* maximizes revenue rate $\xi_t(x_t, r_t) = \sum_{i=1}^n \lambda P_t^i[r_t^i - \Delta^i \pi_{t-1}(x_t)]$. We find r_t^* in five steps.

Step 1. Instead of considering $\mathbb{R}_+^n$, we consider a price space of $\mathbb{R}^n$. By $\mathbb{R}_+^n$ we mean the nonnegative real line not including $+\infty$; by $\mathbb{R}^n$ we mean the real line not including $\pm\infty$. Focusing on the price space $\mathbb{R}^n$ does not affect the solution of $\max \xi_t(x_t, r_t)$ because the optimal r_t^*, if it exists, must be in $\mathbb{R}_+^n$, which can be shown easily. To see this, note that $\Delta^i \pi_{t-1}(x_t) \geq 0$. If $r_t^i \leq \Delta^i \pi_{t-1}(x_t)$ for some i, then we can improve $\xi_t(x_t, r_t)$ by letting $r_t^i = \Delta^i \pi_{t-1}(x_t)$, because increasing $[r_t^i - \Delta^i \pi_{t-1}(x_t)]$ from negative to 0 removes the negative contribution from variate i and increases P_t^j for $j \neq i$ (straightforward from the property of the MNL model).

The reason that infinity value, $+\infty$, is also excluded from our consideration is that setting $r_t^j = +\infty$ for some $j \in S_t$ is suboptimal. Suppose $r_t^j = +\infty$ for some j, and $r_i < +\infty$ for $i \neq j$ and $i \in S_t$. It implies that demand for variate j is "turned off," $P_t^j = 0$, and revenue rate from variate j is zero, $r_t^j P_t^j = 0$. We can show that by setting $r_t^j = M$ and $M > \Delta^j \pi_{t-1}(x_t) + \sum_{i \neq j} r_t^i P_t^i [r_t^i - \Delta^i \pi_{t-1}(x_t)]$ while keeping r_t^i unchanged, we can improve the revenue rate by a positive amount. Let $\{\tilde{P}_t^1, \ldots, \tilde{P}_t^n, \tilde{P}_t^0\}$ be the corresponding sales probabilities. Specifically, the improvement in revenue rate is

$$\begin{aligned}
&\lambda \tilde{P}_t^j [M - \Delta^j \pi_{t-1}(x_t)] + \sum_{i \neq j} \lambda \tilde{P}_t^i [r_t^i - \Delta^i \pi_{t-1}(x_t)] \\
&\quad - \sum_{i \neq j} \lambda P_t^i [r_t^i - \Delta^i \pi_{t-1}(x_t)] \\
&\qquad = \lambda \tilde{P}_t^j [M - \Delta^j \pi_{t-1}(x_t)] - \sum_{i \neq j} \lambda [r_t^i - \Delta^i \pi_{t-1}(x_t)][P_t^i - \tilde{P}_t^i]
\end{aligned}$$

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$$= \lambda \tilde{P}_t^j [M - \Delta^j \pi_{t-1}(x_t)] - \sum_{i \neq j} \lambda [r_t^i - \Delta^i \pi_{t-1}(x_t)] P_t^i \tilde{P}_t^j$$

$$= \lambda \tilde{P}_t^j \left(M - \Delta^j \pi_{t-1}(x_t) - \sum_{i \neq j} r_t^i P_t^i [r_t^i - \Delta^i \pi_{t-1}(x_t)] \right) > 0,$$

where the second equality follows from the identity $P_t^i - \tilde{P}_t^i = P_t^i \tilde{P}_t^j$, $i \neq j$ (straightforward from MNL model).

Step 2. We establish one-to-one mapping between the expanded price space $\mathbb{R}^n$ and the sales probability space $\Psi \doteq \{P = \{P^1, \ldots, P^n, P^0\} \in (0,1)^{n+1}, \sum_{i=0}^n P^i = 1\}$.

It is straightforward that the MNL model below provides a mapping from $\mathbb{R}^n$ to Ψ.

$$P_t^i = \frac{\exp((a_i - r_t^i)/\mu)}{\sum_{j=1}^n \exp((a_j - r_t^j)/\mu) + \exp(u_0/\mu)}, \quad i = 1, 2, \ldots, n, \tag{16}$$

$$P_t^0 = \frac{\exp(u_0/\mu)}{\sum_{j=1}^n \exp((a_j - r_t^j)/\mu) + \exp(u_0/\mu)}. \tag{17}$$

Specifically, for any $r_t \in \mathbb{R}^n$, P_t obtained from (16) and (17) satisfies $P_t^i \in (0,1)$ for all $i = 0, 1, \ldots, n$. We note that $P_t^i = 0$ or 1, $i = 0, 1, \ldots, n$, are excluded from Ψ, as they hold if and only if $\exp((a_i - r_t^i)/\mu) = 0$ or $+\infty$ ($r_t^i = \infty$ or $a_i = \infty$ or $r_t^i = -\infty$), which cannot hold because $r_t \in \mathbb{R}^n$. Therefore, $\forall\, r_t \in \mathbb{R}^n$, (16) and (17) define a corresponding sales probability vector $P_t \in \Psi$. This is an important observation, because it guarantees that if an optimal r_t^* exists, it corresponds to a $P_t^* \in \Psi$.

The following expression provides a mapping from Ψ to $\mathbb{R}^n$.

$$r_t^i = a_i - u_0 - \mu \ln P_t^i + \mu \ln P_t^0, \quad i = 1, \ldots, n. \tag{18}$$

As $P_t^i \in (0,1)$, $i = 0, 1, \ldots, n$, for any $P_t \in \Psi$, we have $-\ln P_t^i \in (0, \infty)$, and r_t^i obtained from (18) belongs to $(-\infty, \infty)$, $i = 1, 2, \ldots, n$. Thus, $\forall\, P_t \in \Psi$, (18) defines a corresponding n-dimensional vector $r_t \in \mathbb{R}^n$. The implication of this observation is that if an optimal P_t^* exists, it corresponds to a $r_t^* \in \mathbb{R}^n$.

Step 3. We prove that $\xi_t(x_t, P_t) \doteq \sum_{i \in S_t} \lambda P_t^i [a_i - u_0 - \mu \ln P_t^i + \mu \ln P_t^0 - \Delta^i \pi_{t-1}(x_t)]$ is concave in P_t for $P_t \in (0,1)^{n+1}$.

The gradient of ξ_t is given by

$$\frac{\partial \xi_t}{\partial P_t^i} = \lambda \left[a_i - u_0 - \mu \ln P_t^i + \mu \ln P_t^0 - \Delta^i \pi_{t-1}(x_t) \right] - \lambda \mu, \quad i = 1, \ldots, n,$$

$$\frac{\partial \xi_t}{\partial P_t^0} = \lambda \mu \left(\sum_{i=1}^n P_t^i \right) \Big/ P_t^0.$$

Because

$$\frac{\partial^2 \xi_t}{\partial P_t^i \partial P_t^i} = -\lambda \mu / P_t^i, \quad i = 1, \ldots, n,$$

$$\frac{\partial^2 \xi_t}{\partial P_t^i \partial P_t^j} = 0, \quad j \neq i,$$

$$\frac{\partial^2 \xi_t}{\partial P_t^0 \partial P_t^0} = -\lambda \mu \left(\sum_{i=1}^n P_t^i \right) \Big/ (P_t^0)^2,$$

$$\frac{\partial^2 \xi_t}{\partial P_t^0 \partial P_t^i} = \lambda \mu / P_t^0, \quad i = 1, \ldots, n,$$

the Hessian of $-\xi_t$ is

$$H_{-\xi_t} = (\lambda \mu)^{n+1} \begin{pmatrix} 1/P_t^1 & 0 & \cdots & 0 & -1/P_t^0 \\ 0 & 1/P_t^2 & \cdots & 0 & -1/P_t^0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1/P_t^n & -1/P_t^0 \\ -1/P_t^0 & -1/P_t^0 & \cdots & -1/P_t^0 & (\sum_{i=1}^n P_t^i)/(P_t^0)^2 \end{pmatrix}.$$

Let $y = (y_i)_{i=1}^{n+1}$. We have

$$y H_{-\xi_t} y^T = (\lambda \mu)^{n+1} \left[\sum_{i=1}^n \frac{y_i^2}{P_t^i} + \frac{y_{n+1}^2}{(P_t^0)^2} \left(\sum_{i=1}^n P_t^i \right) \right] \geq 0,$$

with equality holds if and only if $y = 0$. Therefore ξ_t is concave in P_t for $P_t \in (0,1)^{n+1}$.

Step 4. Because Ψ is a convex subset of $(0,1)^{n+1}$, $\xi_t(x_t, P_t)$ is jointly concave in P_t for $P_t \in \Psi$ as well. Thus, we find the optimal $P_t^* \in \Psi$ that maximizes ξ_t by solving the first order condition (FOC) and proving that the stationary point satisfies $\sum_{i=1}^n P_t^i < 1$ and $P_t^i \in (0,1)$. To do so we replace P_t^0 by $1 - \sum_{i=1}^n P_t^i$, and take the gradient of ξ_t,

$$\begin{aligned} \frac{1}{\lambda} \frac{\partial \xi_t}{\partial P_t^j} &= a_j - u_0 - \mu \ln P_t^j + \mu \ln \left(1 - \sum_{i=1}^n P_t^i \right) - \Delta^j \pi_{t-1}(x_t) \\ &\quad + P_t^j \left(-\frac{\mu}{P_t^j} \right) + \sum_{i=1}^n P_t^i \left(-\frac{\mu}{1 - \sum_{i=1}^n P_t^i} \right) \\ &= a_j - u_0 - \mu \ln P_t^j + \mu \ln \left(1 - \sum_{i=1}^n P_t^i \right) \\ &\quad - \Delta^j \pi_{t-1}(x_t) - \frac{\mu}{1 - \sum_{i=1}^n P_t^i}. \end{aligned}$$

Define

$$m_t = \frac{\mu}{1 - \sum_{i=1}^n P_t^{*i}} = \frac{\mu}{P_t^{0*}}. \tag{19}$$

As $\lambda \neq 0$, FOC $\partial \xi_t / \partial P_t^{j*} = 0$ yields

$$P_t^{j*} = \frac{\mu}{m_t} \exp\left(\frac{a_j - \Delta^j \pi_{t-1}(x_t) - u_0 - m_t}{\mu} \right), \quad j \in S_t. \tag{20}$$

Applying (20) to $\sum_{i \in S_t} P_t^{i*} = 1 - P_t^{0*} = 1 - \mu / m_t$ we have

$$1 - \frac{\mu}{m_t} = \frac{\mu}{m_t} \sum_{i=1}^n \exp\left(\frac{a_i - \Delta^i \pi_{t-1}(x_t) - u_0 - m_t}{\mu} \right).$$

Rearranging terms gives

$$\left(\frac{m_t}{\mu} - 1 \right) \exp\left(\frac{m_t + u_0}{\mu} \right) - \sum_{i=1}^n \exp\left(\frac{a_i - \Delta^i \pi_{t-1}(x_t)}{\mu} \right) = 0. \tag{21}$$

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We claim that for any finite a_is, $\Delta^i \pi_t(x_{t-1})$s, u_0, and μ, there exists one and only one m_t for which (21) holds. Let $\sum$ denote the summation term in (21), which is a positive constant. Define $h(x) = (x/\mu - 1)\exp((x+u_0)/\mu) - \sum$. We have $h(\mu) < 0$, $h(\infty) = \infty$, and $h'(x) = (x/\mu)\exp((x+u_0)/\mu) > 0$ for all $x > 0$. So, there exists a unique $x \in (\mu, \infty)$ satisfying $h(x) = 0$, and thus m_t exists and is unique, and $m_t \in (\mu, \infty)$. By (19), $m_t \in (\mu, \infty)$ means $P_t^{0*} = (1 - \sum_{i=1}^n P_t^{i*}) \in (0,1)$. As all the P_t^{i*}s are positive by (20), $(1 - \sum_{i=1}^n P_t^{i*}) \in (0,1)$ implies $P_t^{i*} \in (0,1)$. Therefore, (19), (20), and (21) provide the unique optimal probability vector $P_t^* \in \Psi$.

Step 5. We now translate the optimal P_t^* to r_t^* and show that $r_t^* \in \mathbb{R}_+^n$.

Substituting (19) and (20) into (18) gives

$$r_t^{j*} = \Delta^j \pi_{t-1}(x_t) + m_t > 0$$

because $\Delta^j \pi_{t-1}(x_t) \geq 0$ and $m_t > 0$.

As r_t^* and P_t^* maximize ξ_t, by substituting them in

$$\pi_t(x_t) = \xi_t(x_t, r_t^*) + \pi_{t-1}(x_t), \tag{22}$$

we have

$$\begin{aligned}\pi_t(x_t) &= \sum_{i\in S_t} \lambda P_t^{i*}[r_t^{i*} - \Delta^i \pi_{t-1}(x_t)] + \pi_{t-1}(x_t)\\ &= \lambda m_t \sum_{i\in S_t} P_t^{i*} + \pi_{t-1}(x_t)\\ &= \lambda m_t (1 - P_t^{0*}) + \pi_{t-1}(x_t)\\ &= \lambda(m_t - \mu) + \pi_{t-1}(x_t).\end{aligned}$$

Applying the above equality recursively, and the terminal condition $\pi_0(x_t) = 0$, we obtain

$$\pi_t(x_t) = \lambda \sum_{s=1}^{t} [m_s(x_t) - \mu]. \quad \square$$

Proof of Corollary 2. Through backward induction we can prove that when $x \geq t$, $\Delta^i \pi_s(x) = 0$ for $s \leq t$. Hence, the first order condition (10) for any $s \leq t$ becomes (12). Therefore, $m_s = m$ for $s \leq t$. From (19), $P_s^{0*} = \mu/m$. Then, from (20), $P_s^{i*} = (\mu/m)\exp((a_i - m - u_0)/\mu)$. $\square$

Proof of Proposition 1. Now the optimality equation (5) becomes

$$\pi_t(x_t) = \max_{r_t \in \mathbf{R}_+} \sum_{i\in S_t} \lambda P_t^i [r_t - \Delta^i \pi_{t-1}(x_t)] + \pi_{t-1}(x_t). \tag{23}$$

Define

$$\xi_t^u(x_t, r_t) = \sum_{i\in S_t} \lambda P_t^i [r_t - \Delta^i \pi_{t-1}(x_t)]. \tag{24}$$

Then the optimal price is the maximizer of $\xi_t^u(x_t, r_t)$. It is easy to verify that $\partial P_t^i/\partial r_t = -(1/\mu)P_t^i P_t^0$ and $\partial P_t^0/\partial r_t = (1/\mu)P_t^0(1 - P_t^0)$. So

$$\begin{aligned}\frac{\partial \xi_t^u}{\partial r_t} &= \lambda \sum_{i\in S_t}\left(P_t^i - \frac{1}{\mu} P_t^i P_t^0 \big(r_t - \Delta^i \pi_{t-1}(x_t)\big)\right)\\ &= \lambda(1 - P_t^0) - \frac{1}{\mu} P_t^0 \xi_t^u,\\ \frac{\partial^2 \xi_t^u}{\partial r_t^2} &= -\frac{\lambda}{\mu} P_t^0 (1 - P_t^0) - \frac{1}{\mu^2} P_t^0 (1 - P_t^0)\xi_t^u - \frac{1}{\mu} P_t^0 \frac{\partial \xi_t^u}{\partial r_t}\\ &= -\frac{1}{\mu^2} P_t^0 (1 - P_t^0)(\lambda\mu + \xi_t^u) - \frac{1}{\mu} P_t^0 \frac{\partial \xi_t^u}{\partial r_t}.\end{aligned}$$

The FOC gives

$$P_t^0 = \frac{1}{\xi_t^u/(\lambda\mu) + 1} \tag{25}$$

$$\xi_t^u = \lambda\mu\left(\frac{1}{P_t^{0*}} - 1\right). \tag{26}$$

From (24), any nonpositive ξ_t^u can be improved by increasing r_t so that $r_t > \Delta^i \pi_{t-1}(x_t)$ and $P_t^i > 0$ for all $i \in S_t$. So we have $\xi_t^u > 0$ and $0 < P_t^0 < 1$ (by (25)) at any stationary point.

Evaluated at any stationary point, the second-order derivative is

$$\frac{\partial^2 \xi_t^u}{\partial r_t^2} = -\frac{\lambda}{\mu}(1 - P_t^0),$$

which is always negative. Therefore, ξ_t^u is strictly quasi-concave in r_t and the solution of the FOC (26) is unique and is the global maximizer of ξ_t^u.

We use the same notation as in Theorem 1, i.e., $m_t = \mu/P_t^{0*}$. By (2) we have

$$\exp\left(-\frac{r_t^* + u_0}{\mu}\right) = \left(\frac{m_t}{\mu} - 1\right) \sum_{i\in S_t} \exp\left(-\frac{a_i}{\mu}\right). \tag{27}$$

As $P_t^{i*} = P_t^{0*}\exp((a_i - r_t^* - u_0)/\mu)$, (24) plus (26) gives

$$\begin{aligned}&r_t^*\left(1 - \frac{\mu}{m_t}\right) - \frac{\mu}{m_t}\exp\left(-\frac{r_t^* + u_0}{\mu}\right)\\ &\quad\cdot \sum_{i\in S_t}\left(\exp\left(\frac{a^i}{\mu}\right)\Delta^i \pi_{t-1}(x_t)\right) = m_t - \mu.\end{aligned} \tag{28}$$

Substituting (27) into (28) gives

$$r_t^* = \frac{\sum_{i\in S_t}\exp(a^i/\mu)\Delta^i \pi_{t-1}(x_t)}{\sum_{i\in S_t}\exp(a^i/\mu)} + m_t = \Delta^w \pi_{t-1}(x_t) + m_t.$$

Defining $a_w = \mu \ln(\sum_{i\in S_t}\exp(a_i/\mu))$ and substituting the above expression of r_t^* back to (27), we have

$$\begin{aligned}\left(\frac{m_t}{\mu} - 1\right)\exp\left(\frac{m_t + u_0}{\mu}\right) &= \exp\left(-\frac{\Delta^w \pi_{t-1}(x_t)}{\mu}\right)\sum_{i\in S_t}\exp\left(\frac{a_i}{\mu}\right)\\ &= \exp\left(\frac{a_w - \Delta^w \pi_{t-1}(x_t)}{\mu}\right).\end{aligned}$$

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The uniqueness of the solution to the above equation is shown in the proof of Theorem 1.

Finally, from (23)–(26) and the boundary condition (4), we have

$$\pi_t(x_t) = \lambda \sum_{s=1}^{t} [m_s(x_t) - \mu]. \quad \square$$

Proof of Proposition 2. Now the optimality equation (5) becomes

$$\pi_t(x_t) = \max_{r_t \in \mathbf{R}_+^{n_1}} \Bigg(\sum_{i \in S_{t1}} \lambda P_t^i [r_t^i - \Delta^i \pi_{t-1}(x_t)] + \sum_{j \in S_{t2}} \lambda P_t^j [r^j - \Delta^j \pi_{t-1}(x_t)] \Bigg) + \pi_{t-1}(x_t). \tag{29}$$

Let $P_{t1} \in (0,1)^{n_1}$ denote the probability vector of the variates in the set S_{t1}. We note that $0 < P_t^i < 1$ for all $i \in S_{t1}$, for the same reason as discussed in the proof of Theorem 1. Similarly, we have $\sum_{i \in S_{t1}} P_t^i < 1$. Define $\theta_j = \exp((a_j - r^j - u_0)/\mu)$ and $\Theta = \sum_{j \in S_{t2}} \theta_j$. From (1) and (2), $P_t^j = \theta_j P_t^0$ for all $j \in S_{t2}$. As $\sum_{i \in S_{t1}} P_t^i + \sum_{j \in S_{t2}} P_t^j + P_t^0 = 1$, we have

$$P_t^0 = \frac{1}{1+\Theta}\Bigg(1 - \sum_{i \in S_{t1}} P_t^i\Bigg), \quad P_t^j = \frac{\theta_j}{1+\Theta}\Bigg(1 - \sum_{i \in S_{t1}} P_t^i\Bigg). \tag{30}$$

Then the optimality equation (29) can be rewritten as

$$\pi_t(x_t) = \max_{P_{t1} \in \mathbf{R}_+^{n_1}} \Bigg(\sum_{i \in S_{t1}} \lambda P_t^i [a_i - u_0 - \mu \ln P_t^i + \mu \ln P_t^0 - \Delta^i \pi_{t-1}(x_t)] + \sum_{j \in S_{t2}} \lambda P_t^j [r^j - \Delta^j \pi_{t-1}(x_t)] \Bigg) + \pi_{t-1}(x_t).$$

Define

$$\xi_t^m(x_t, P_{t1}) = \sum_{i \in S_{t1}} \lambda P_t^i [a_i - u_0 - \mu \ln P_t^i + \mu \ln P_t^0 - \Delta^i \pi_{t-1}(x_t)] + \sum_{j \in S_{t2}} \lambda P_t^j [r^j - \Delta^j \pi_{t-1}(x_t)]. \tag{31}$$

From (30), $\partial P_t^0 / \partial P_t^i = -1/(1+\Theta)$ and $\partial P_t^j / \partial P_t^i = -\theta_j/(1+\Theta)$ for all $i \in S_{t1}$ and $j \in S_{t2}$. Also, $\sum_{i \in S_{t1}} P_t^i = 1 - (1+\Theta) P_t^0$. Therefore,

$$\begin{aligned}
\frac{1}{\lambda} \frac{\partial \xi_t^m}{\partial P_t^k} &= a_k - u_0 - \mu \ln P_t^k + \mu \ln P_t^0 - \Delta^k \pi_{t-1}(x_t) \\
&\quad + P_t^k \Bigg(-\frac{\mu}{P_t^k} - \frac{\mu}{(1+\Theta) P_t^0} \Bigg) + \sum_{i \in S_{t1} \setminus \{k\}} P_t^i \Bigg(-\frac{\mu}{(1+\Theta) P_t^0} \Bigg) \\
&\quad - \frac{1}{1+\Theta} \sum_{j \in S_{t2}} \theta_j [r^j - \Delta^j \pi_{t-1}(x_t)] \\
&= a_k - u_0 - \mu \ln P_t^k + \mu \ln P_t^0 - \Delta^k \pi_{t-1}(x_t) - \frac{\mu}{(1+\Theta) P_t^0} \\
&\quad - \frac{1}{1+\Theta} \sum_{j \in S_{t2}} \theta_j [r^j - \Delta^j \pi_{t-1}(x_t)].
\end{aligned}$$

Then

$$\frac{\partial^2 \xi_t^m}{\partial P_t^{k\,2}} = -\lambda\mu \Bigg(\frac{1}{P_t^k} + \frac{1}{(1+\Theta) P_t^0} + \frac{1}{(1+\Theta)^2 P_t^{0\,2}} \Bigg),$$

$$\frac{\partial^2 \xi_t^m}{\partial P_t^k \partial P_t^l} = -\lambda\mu \Bigg(\frac{1}{(1+\Theta) P_t^0} + \frac{1}{(1+\Theta)^2 P_t^{0\,2}} \Bigg).$$

Define $\rho_i = 1/P_t^i$ for $i = 1, \ldots, n$ and $\rho_0 = 1/((1+\Theta)P_t^0) + 1/((1+\Theta)^2 P_t^{0\,2})$. Note that $\rho_i > 0$ for all $i = 0, \ldots, n$. The Hessian of $-\xi_t^m$ is

$$H_{-\xi_t^m} = (\lambda\mu)^n \begin{pmatrix} \rho_1 + \rho_0 & \rho_0 & \cdots & \rho_0 \\ \rho_0 & \rho_2 + \rho_0 & \cdots & \rho_0 \\ \vdots & \vdots & \ddots & \vdots \\ \rho_0 & \rho_0 & \cdots & \rho_n + \rho_0 \end{pmatrix}.$$

Let $y = (y_j)_{j=1}^n$, we have $y H_{-\xi_t^m} y^T = (\lambda\mu)^n (\sum_{i=1}^n \rho_i y_i^2 + \rho_0 (\sum_{i=1}^n y_i)^2) \geq 0$, and the equality holds if and only if $y = 0$. So $H_{-\xi_t^m}$ is positive definite and $\xi_t^m(x_t, P_t)$ is strictly joint concave in P_t, and thus, the optimal probability vector P_{t1} is uniquely given by the FOCs.

Define

$$m_t = \frac{\mu}{(1+\Theta) P_t^{0*}} + \frac{1}{1+\Theta} \sum_{j \in S_{t2}} \theta_j [r^j - \Delta^j \pi_{t-1}(x_t)]. \tag{32}$$

From the FOCs we have

$$P_t^{k*} = P_t^{0*} \exp\Bigg(\frac{a_k - \Delta^k \pi_{t-1}(x_t) - m_t - u_0}{\mu} \Bigg). \tag{33}$$

As $P_t^{k*} = P_t^{0*} \exp((a_k - r_t^{k*} - u_0)/\mu)$, (33) gives

$$r_t^{k*} = \Delta^k \pi_{t-1}(x_t) + m_t. \tag{34}$$

Recall $\sum_{i \in S_{t1}} P_t^i + \sum_{j \in S_{t2}} P_t^j + P_t^0 = 1$. From (2),

$$\frac{1}{P_t^{0*}} = 1 + \Theta + \sum_{i \in S_{t1}} \exp\Bigg(\frac{a_i - r_t^{i*} - u_0}{\mu} \Bigg). \tag{35}$$

Substituting (34) and (35) into (32) gives

$$(1+\Theta) m_t = \mu \Bigg(1 + \Theta + \sum_{i \in S_{t1}} \exp\Bigg(\frac{a_i - \Delta^i \pi_{t-1}(x_t) - m_t - u_0}{\mu} \Bigg) \Bigg) + \sum_{j \in S_{t2}} \theta_j [r^j - \Delta^j \pi_{t-1}(x_t)].$$

After rearranging the terms we have

$$\Bigg((1+\Theta) \Bigg(\frac{m_t}{\mu} - 1 \Bigg) - \frac{1}{\mu} \sum_{j \in S_{t2}} \theta_j [r^j - \Delta^j \pi_t(x_t)] \Bigg) \exp\Bigg(\frac{m_t + u_0}{\mu} \Bigg) = \sum_{i \in S_{t1}} \exp\Bigg(\frac{a_i - \Delta^i \pi_{t-1}(x_t)}{\mu} \Bigg). \tag{36}$$

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Substituting $\theta_j = \exp((a_j - r^j - u_0)/\mu)$ and $\Theta = \sum_{j\in S_{t2}} \theta_j$ into (36) gives

$$\left(\frac{m_t}{\mu} - 1\right)\exp\left(\frac{m_t + u_0}{\mu}\right) + \sum_{j\in S_{t2}}\left(\left(\frac{m_t - r^j + \Delta^j \pi_{t-1}(x_t)}{\mu} - 1\right)\exp\left(\frac{m_t + a_j - r^j}{\mu}\right)\right) = \sum_{i\in S_{t1}} \exp\left(\frac{a_i - \Delta^i \pi_{t-1}(x_t)}{\mu}\right). \quad (37)$$

Defining $S_2^+(x_t) = S_2(x_t) \cup \{0\}$, $v_0 = 0$, $v_j = r^j - \Delta^j \pi_{t-1}(x_t)$, and $u_j = a_j - r^j$ for all $j \in S_2(x_t)$, we have (15).

As both Θ and the summation over S_{t2} are constant, the uniqueness of the solution to (36), which is equivalent to (15), can be shown in the same way as in the proof of Theorem 1.

Recall $r_t^i = a_i - u_0 - \mu \ln P_t^i + \mu \ln P_t^0$ and $P_t^j = \theta_j P_t^0$ for all $j \in S_{t2}$. From (31), (32), and (34),

$$\begin{aligned}\pi_t(x_t) - \pi_{t-1}(x_t) &= \lambda\left(m_t(1 - (1+\Theta)P_t^{0*}) + P_t^{0*}\sum_{j\in S_{t2}} \theta_j[r^j - \Delta^j \pi_{t-1}(x_t)]\right)\\ &= \lambda(m_t - \mu).\end{aligned}$$

Taking into account the boundary (4), we have

$$\pi_t(x_t) = \lambda\sum_{s=1}^{t}[m_s(x_t) - \mu]. \quad \square$$

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